

IDAR-OBERSTEIN, Germany

WMO: 106180

Lat: 49.6928N

Lon: 7.3264E

Elev: 376

StdP: 96.89

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.1	-12.9	1.3	-8.9	-10.4	1.6	-6.5	13.4	6.8	12.1	6.5	1.9	60	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	29.8	19.0	27.6	18.2	25.7	17.4	19.9	27.5	19.0	25.8	18.1	24.2	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	13.0	21.9	16.6	12.3	21.2	15.8	11.7	20.5	58.4	27.6	55.4	25.8	52.5	24.1	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.2	8.6	7.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-14.9	33.0	3.1	1.7	-17.1	34.2	-18.9	35.2	-20.7	36.1	-22.9	37.3
			-14.8	21.5	3.0	0.8	-17.0	22.1	-18.8	22.5	-20.5	23.0	-22.7	23.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	0.0	1.6	4.6	8.6	12.5	15.8	17.4	17.3	13.5	9.2	4.7	1.0
	DBStd	7.22	4.62	4.28	3.88	3.84	3.61	3.47	3.28	3.14	3.03	3.73	3.58	4.36
	HDD10.0	1309	309	236	172	71	17	1	0	0	4	61	161	278
	HDD18.3	3521	567	469	427	292	183	92	58	57	147	284	408	537
	CDD10.0	904	0	0	3	29	94	174	230	227	109	35	2	0
	CDD18.3	75	0	0	0	0	3	15	30	25	2	0	0	0
	CDH23.3	1017	0	0	0	5	54	193	376	338	49	1	0	0
	CDH26.7	286	0	0	0	0	7	42	115	114	7	0	0	0
Wind	WSAvg	3.1	3.5	3.8	3.5	3.0	3.0	2.8	3.0	2.7	2.7	2.9	3.0	3.6
Precipitation	PrecAvg	796	76	66	58	47	62	64	67	68	59	68	72	88
	PrecMax	1090	234	208	176	131	146	148	196	156	126	233	149	257
	PrecMin	547	11	6	10	3	12	21	26	8	17	14	6	6
	PrecStd	135	43	45	39	26	30	35	37	35	29	42	34	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	12.3	18.4	23.9	27.6	30.3	32.1	34.1	27.8	21.9	14.5	11.0
		MCWB	9.5	7.9	10.3	13.5	17.7	19.2	19.9	19.3	17.7	16.1	11.8	9.7
	2%	DB	9.6	10.2	15.4	21.3	24.7	27.7	29.8	29.7	24.6	18.5	12.3	9.8
		MCWB	8.5	7.2	9.2	12.7	16.1	18.4	19.2	18.8	16.9	14.3	10.4	8.6
	5%	DB	8.1	8.6	12.9	18.7	22.6	25.5	27.6	27.0	21.9	16.4	10.9	8.5
		MCWB	6.9	6.7	8.3	11.4	14.7	17.3	18.3	17.8	15.7	13.3	9.6	7.4
	10%	DB	6.6	7.4	10.8	16.3	20.3	23.5	25.2	24.7	19.6	14.7	9.6	7.2
		MCWB	5.5	5.9	7.8	10.2	13.5	16.5	17.4	17.1	14.6	12.5	8.4	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	9.0	11.5	14.6	19.0	20.4	21.1	20.7	19.0	16.7	12.2	10.1
		MCDB	10.5	10.6	16.1	22.5	25.4	27.5	29.3	29.5	24.8	20.1	13.7	10.7
	2%	WB	8.5	7.9	10.2	13.1	17.1	19.3	19.9	19.7	17.6	15.1	10.8	8.7
		MCDB	9.5	9.4	13.8	19.6	23.5	25.9	27.8	27.3	22.9	17.8	12.0	9.6
	5%	WB	6.9	7.0	9.1	11.9	15.5	18.2	18.9	18.6	16.4	13.8	9.7	7.6
		MCDB	7.9	8.3	12.2	17.3	21.0	24.2	25.9	25.2	20.9	15.8	10.7	8.4
	10%	WB	5.5	5.9	8.0	10.8	14.1	17.1	17.9	17.7	15.2	12.6	8.5	6.2
		MCDB	6.5	7.4	10.6	15.5	18.9	22.4	23.6	23.6	18.7	14.5	9.4	7.1
Mean Daily Temperature Range	5% DB	MDBR	4.8	6.1	8.3	10.5	10.8	11.3	11.1	11.3	10.4	8.1	5.2	4.4
		MCDBR	5.5	8.2	12.7	15.4	15.4	15.4	16.0	16.8	14.9	10.8	7.2	4.9
		MCWBR	4.7	5.7	7.6	7.7	7.6	7.3	6.8	7.2	7.7	6.8	5.3	4.2
	5% WB	MCDBR	5.3	6.6	10.7	13.3	13.8	13.8	14.3	13.9	12.4	9.5	6.3	4.9
		MCWBR	4.8	4.9	6.7	6.9	7.1	6.9	6.5	6.5	6.9	6.1	5.0	4.4
Clear-Sky Solar Irradiance	taub	0.309	0.336	0.367	0.393	0.400	0.403	0.401	0.397	0.371	0.358	0.331	0.307	
	taud	2.418	2.364	2.295	2.266	2.271	2.289	2.284	2.323	2.377	2.401	2.415	2.410	
	Ebn at Noon	740	801	834	850	858	856	851	837	823	764	701	684	
	Edh at Noon	64	85	108	123	128	127	125	115	98	81	64	56	
All-Sky Solar Radiation	RadAvg	0.80	1.53	2.71	4.25	5.02	5.59	5.30	4.54	3.41	1.96	0.90	0.62	
	RadStd	0.12	0.29	0.41	0.53	0.52	0.50	0.56	0.47	0.41	0.18	0.10	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page